

SUJITH MAKAM

AI Developer | Software Engineer

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Education

Indian Institute of Information Technology (IIIT), Dharwad

DEC 2021 - JULY 2025

B.Tech in Computer Science and Engineering

CGPA: 8.41

Technical Skills

AI/ML: Conversational AI, Generative AI, Agentic AI, LLMs, RAG, Prompt Engineering, AI Governance, Multi-Agent Systems

AI Frameworks: Cognigy, Azure AI Foundry, Azure OpenAI, CrewAI, Google ADK, Langfuse, AgentOps

Languages: Python, TypeScript, JavaScript, C++, C

Cloud & DevOps: AWS, Azure, AWS Lambda, Azure Functions, AWS CDK, Docker

Web: ReactJS, Node.js, Express.js

Databases: MongoDB, MySQL, Vector Databases

Operating Systems: Ubuntu, Windows

Experience

CGI

Jul. 2025 - Present

Software Engineer

- Architected **omnichannel Conversational AI solutions** on Cognigy and Genesys Cloud, integrating Azure AI Foundry and AWS services to automate multi-channel (voice, chat) customer support workflows.
- Built production-ready LLM applications including AI-powered call summarization and automated QA evaluation using Azure OpenAI, RAG pipelines and enterprise knowledge bases.
- Designed **multi-agent conversational** orchestration workflows enabling autonomous, context-aware task execution across **customer support** conversations, improving containment and SLA compliance.
- Implemented **end-to-end Infrastructure-as-Code (IaC)** using **AWS CDK (TypeScript)** to provision and deploy complete **Amazon Connect** environments, enabling scalable, automated cloud contact-center deployments.

Indian Institute of Science Bangalore

May. 2024 - July. 2024

Research Intern

- Investigated LLM Pruning techniques for improving model efficiency, optimizing the attention and feed-forward layers of GPT-2. Achieved a 5.2% reduction in instruction cache misses through tiled matrix multiplication implementation.
- Conducted detailed profiling of GPT-2 using Intel VTune, identifying bottlenecks and enhancing cache efficiency, which resulted in a 4x speedup in attention layer execution.
- Optimized identified bottlenecks in the attention and feed-forward layers of GPT-2 and Reduced memory bandwidth usage by 34.8 %, decreasing memory-bound issues and ensuring more efficient execution
- Implemented custom classes (e.g., tiled matrix multiplication) for performance improvement in the attention layer.

Projects

AI Governance Platform



- Developed an enterprise AI Governance Platform supporting multi-agent workflows using Azure AI Foundry, Google ADK, CrewAI and Agent Gateway for secure orchestration of LLM agents.
- Integrated Langfuse and AgentOps for end-to-end agent observability, prompt tracing, KPI monitoring, latency analysis and production evaluation.
- Implemented Responsible AI controls including prompt injection prevention, sensitive data masking, virtual API isolation, policy enforcement and secure tool execution.

Automated QA Evaluation



- Designed and deployed an LLM-powered QA Evaluation Platform using Azure AI Foundry, Azure Open AI, Azure Functions, API Gateway and REST APIs to automatically evaluate customer support calls against predefined quality metrics.
- Built RAG pipelines over enterprise Knowledge Base documents using vector search and embeddings, enabling context-aware quality assessment and reducing hallucinations.
- Automated quality evaluation for **40,000+ customer calls with 99% evaluation coverage**, enabling scalable AI-driven QA, reducing manual review effort and accelerating agent feedback cycles.

Lightweight IoT Intrusion Detection System (Edge-AI)



- Developed a scalable Curriculum Learning framework with XAI (LIME) validation, utilizing optimized GRU/LSTM attention layers to progressively detect complex cyber-attacks.
- Attained 98 % accuracy on the CIC-IoV-2024 dataset while reducing model size to 367KB via quantization and pruning, enabling high-efficiency deployment on resource-constrained edge devices.

FAPL-DM-BC: Blockchain-Federated Security Framework



- Designed a privacy-preserving IoV framework merging Federated Learning with Blockchain smart contracts for decentralized model validation and tamper-proof provenance.
- Achieved 93.1% accuracy with 42ms latency by engineering a Dockerized microservices architecture (using SMPC) and integrating a dual-model XAI feedback loop (SHAP/LIME) for real-time threat detection.

Certifications

- Full Stack Web Development from Udemy.
- Machine Learning A-Z from Udemy.



Research Publications

Research Paper Publication and Presentation: COMSNETS 2026

Published and presented a paper titled "FAPL-DM-BC: Adaptive Privacy-Aware, Blockchain-Backed FL Framework with Dual-Model XAI for the IoVs" at the COMputer COMMunication Systems and NETworkS Forum 2026

Research Paper Publication and Presentation: IEEE FNWF 2025

Published and presented a paper titled "Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI" at the IEEE Future Networks World Forum 2025

Research Paper Publication and Presentation: IEEE FNWF 2025

Published and presented a paper titled FedGuard-IIoT: A Scalable Federated Learning Architecture for Network-IDS in the Industrial IoT" at the IEEE Future Networks World Forum 2025

Research Paper Publication and Presentation: IEEE SPACE 2025

Published and presented a paper titled "Predicting Stellar Metallicity: A Comparative Analysis of Regression Models for Solar Twin Stars" at the IEEE International Conference on Signal Processing, Computing and Control (SPACE) 2025

Achievements

TCS TechBytes: Dharwad Regional Finals Runner Up

A Prestigious Tech Quiz conducted by TCS and BITES in Karnataka

State Rank Holder: Andhra Pradesh State Board Topper

Scored 990/1000 in my state board examinations